SPEAR-5G: A Stackelberg Game-Theoretic Framework for Attack-Resilient Deployment of Non-Public 5G Networks

***Abstract*—Enterprises are increasingly turning to non-public 5G networks to obtain assured service-level agreements, mobility, and infrastructure control that public mobile networks cannot guarantee. Choosing among candidate deployment architectures – physical or cloud-native network functions, standalone or shared radio access, and dedicated network slices – involves a fundamental trade-off between deployment cost and security exposure. Existing cost-driven approaches to this problem typically treat the adversary as a fixed cost or benefit term, which overlooks the fact that a rational attacker adapts its effort to the specific architecture it observes, investing more against exposed, shared infrastructure and less against isolated deployments. This paper introduces SPEAR-5G, a Stackelberg game-theoretic framework in which the enterprise (leader) commits to a possibly randomized deployment strategy across candidate architectures and an attacker (follower) responds with an optimal attack-effort level. We formalize the resulting game, derive the attacker's best-response effort from a first-order condition, and compute the enterprise's optimal commitment as a Strong Stackelberg Equilibrium (SSE) via a tractable simplex-search procedure. We further extend the model to a Bayesian setting that captures uncertainty about attacker sophistication (opportunistic vs. advanced adversaries). Using a calibrated numerical simulation, we show that SPEAR-5G's equilibrium strategy dominates both a static, deterministic cost-minimizing baseline and a naive uniform-randomization baseline across a range of attacker strengths, reducing the enterprise's total expected cost by up to 19% relative to the static baseline while achieving a strictly better cost–security Pareto frontier.**

***Index Terms*—Non-public networks, 5G, Stackelberg games, security games, core network, game theory, network deployment**

I. Introduction

Enterprises increasingly deploy non-public (private) 5G networks to obtain assured service-level agreements, mobility, and control that public mobile networks and VPN-over-consumer-network alternatives cannot guarantee [1]. A central design question for such deployments is choosing among candidate deployment architectures – physical vs. cloud-native network functions; standalone vs. shared RAN and core; dedicated slices – each with a different balance of deployment cost and attack surface. A natural way to approach this question is as a constrained optimization: minimize deployment cost, maximize the cost incurred by an attacker, and minimize the attacker's benefit, as proposed for non-public 5G networks by Tripathi *et al.* [2]. Such cost-driven formulations are a valuable starting point, but they share a structural limitation common to many cost-based security frameworks: the attacker's behavior is folded into a *fixed* cost/benefit term rather than modeled as a *best response* to the enterprise's observable (or partially observable) deployment choice. In practice, a rational adversary adapts its attack effort to the architecture it faces – investing more against exposed, publicly-shared infrastructure and less against isolated, dedicated deployments. Ignoring this strategic adaptation can lead an enterprise to systematically under- or over-invest in security.

We address this gap by casting non-public 5G deployment selection as a two-player *Stackelberg security game* [3], [4]. The enterprise (defender/leader) commits to a possibly randomized deployment policy over a set of candidate architectures; the attacker (follower) observes this policy (or its distribution, e.g. across multiple enterprise branches/services) and selects an attack effort level that maximizes its own expected utility. We compute the enterprise's optimal commitment as a Strong Stackelberg Equilibrium (SSE), following the methodology used in deployed security-game systems such as ARMOR/IRIS/PROTECT [3], [5], and augment it with a Bayesian layer to capture uncertainty about attacker sophistication. Our specific contributions are:

- We formalize **SPEAR-5G**, a Stackelberg game-theoretic framework for non-public 5G deployment selection (Section III), built on a standard service and attack model whose cost taxonomy (Table I) serves as the payoff primitives of the game (Section IV).
- We derive the attacker's best-response effort in closed form via a first-order condition and describe a tractable simplex-search procedure for computing the enterprise's optimal (Bayesian) Stackelberg strategy (Section IV).
- We provide a simulated evaluation methodology and report four evaluations (Section VI) showing that SPEAR-5G's equilibrium mixed strategy Pareto-dominates both a static, cost-only baseline and a naive uniform-randomization baseline.

II. Related Work

Non-public 5G security. Tripathi *et al.* [2] propose a cost-driven multi-objective optimization for selecting among non-public 5G deployment architectures; we adopt a compatible service model, attack model, and cost taxonomy as the foundation for our game-theoretic formulation. Complementary empirical work examines weaknesses in the security-mode command and NAS/RRC procedures that our attack model abstracts as exposure [6], [7]. ENISA's threat landscape report [8] and Tang *et al.* [9] catalogue 5G core threats end-to-

end, and Ramezanpour *et al.* [10] survey vulnerabilities across 5G/6G and WiFi 6 coexistence; these inform our exposure parameters (Section V). Park *et al.* [11] study control-plane placement trade-offs for private 5G deployments, and Wen *et al.* [1] survey private 5G architectures broadly.

Game theory for network/infrastructure security. Stackelberg security games (SSGs) originate in the counter-terrorism and infrastructure protection literature, where a defender commits to a randomized resource allocation observed by a strategic attacker [3]. Paruchuri *et al.* [5] introduced the DOBSS algorithm for efficiently computing Bayesian Stackelberg equilibria, and Yin *et al.* [4] analyzed the relationship between Stackelberg and Nash solution concepts in security games; both results underpin our equilibrium formulation and Bayesian extension. Recent work continues to apply Stackelberg formulations to cyber-defense problems, including coordinated multi-domain deception [12].

Game theory for 5G resource/slicing decisions. Game-theoretic methods have been used extensively for 5G *resource* allocation: Caballero *et al.* [13] model multi-tenant slice customization as a non-cooperative game with provable convergence, and Datar *et al.* [14] formulate a Stackelberg game for pricing and resource allocation in 5G network slicing. These works target *performance/economic* objectives among cooperative or self-interested tenants, not an adversarial attacker. To our knowledge, no prior work applies a Stackelberg security-game formulation to the specific problem of selecting *among* non-public 5G deployment architectures under an adaptive attacker – this is the gap SPEAR-5G fills.

III. System Model and Attack Model

We build on a service model and attack model consistent with recent work on non-public 5G deployment security [2], summarized here for self-containment.

Service model. The enterprise deploys n network services realized by network functions (NFs) drawn from four deployment architectures: (i) enterprise-owned RAN with physical core NFs; (ii) enterprise-owned RAN with cloud-native core NFs; (iii) shared RAN and shared cloud-native core NFs; and (iv) core NFs realized as a dedicated slice on shared infrastructure. User equipment (UE) connect via 5G (optionally 4G for backward compatibility) and interact with the core over registration/session-management procedures (N2/N3/N6 interfaces).

Attack model. The attacker targets weaknesses in the security-mode command (SMC), which governs UE authentication/registration, to disclose UE identity/location information, and can additionally attempt downgrade attacks affecting PDU sessions [2]. The attacker incurs costs for access, path length, exploit complexity, and risk of detection, and the attack model assumes the enterprise's server-side application logic is out of scope. Table I lists the cost metrics used as payoff primitives throughout this paper.

From static costs to a game. Prior cost-driven formulations treat $C_{res}{+}C_{acc}{+}C_{ap}$ (attack cost) and $C_{ue}{+}C_{nc}{+}C_d$ (attack benefit) as *fixed* quantities used only as feasibility constraints ($C_{res}{+}C_{acc}{+}C_{ap} < C_{ue}{+}C_{nc}{+}C_d$). SPEAR-5G instead makes the attack cost and attack benefit *functions of an attack-effort variable e that the attacker chooses strategically*, and makes the probability that an attack succeeds a function of both e and the deployment architecture's exposure. This is the minimal change needed to convert a static cost optimization into a game.

TABLE I
Cost metrics used as payoff primitives

Metric	Description
C_p, C_{ps}, C_v, C_{vs}	Deployment cost: physical/cloud-native × standalone/shared
C_{res}, C_{acc}, C_{ap}	Attacker's resource, access, and path cost
C_{ue}, C_{nc}, C_d	Cost/benefit of UE, network, and data compromise

IV. Proposed Framework: SPEAR-5G

A. Players, Strategies, and Payoffs

Leader (enterprise). The enterprise's strategy is a probability distribution $\boldsymbol{\sigma} = (\sigma_1, \sigma_2, \sigma_3, \sigma_4)$, $\sigma_j \geq 0$, $\sum_j \sigma_j = 1$, over the four deployment architectures of Section III, interpreted either as (a) a randomized choice for a single enterprise deployment under partial attacker observability, or (b) the empirical mix of architectures across the enterprise's branches/services – both are standard interpretations of a mixed strategy in Stackelberg security games [3]. Each architecture j has a deployment cost $C_j \in \{C_p, C_{ps}, C_v, C_{vs}\}$ and an exposure coefficient θ_j capturing its attack surface (isolated/physical $\Rightarrow$ low θ_j; shared/public infrastructure $\Rightarrow$ high θ_j), consistent with the qualitative trends described in [2]: shared cloud-native deployments are cost-efficient but more exposed, while dedicated slices trade some of that cost efficiency for isolation.

Follower (attacker). The attacker chooses a non-negative effort level $e \geq 0$ (aggregating C_{res}, C_{acc}, C_{ap} from Table I into a single scalar investment) after observing $\boldsymbol{\sigma}$. Given effort e against an architecture with exposure θ_j, the probability that the SMC/downgrade attack succeeds is modeled as a saturating exponential

$$p_j(e) = 1 - e^{-\theta_j e}, \tag{1}$$

consistent with standard diminishing-returns models of attacker investment used in the security-games literature (e.g., [3]). The attacker's expected utility, aggregating C_{ue}, C_{nc}, C_d into a single benefit scale B, is

$$U_A(\boldsymbol{\sigma}, e) = B \sum_{j=1}^{4} \sigma_j \, p_j(e) \; - \; c_1 e^2, \tag{2}$$

where $c_1 > 0$ is the attacker's marginal cost-of-effort coefficient (smaller $c_1 \Rightarrow$ a cheaper/more capable, i.e. stronger, attacker). The enterprise's expected utility combines deployment cost with expected loss (L is the enterprise's loss scale, generally $L \geq B$ to reflect collateral costs not captured by

Algorithm 1 SPEAR-5G Equilibrium Computation

Input: costs $\{C_j\}$, exposures $\{\theta_j\}$, B, L, λ, attacker cost coeff. c_1, simplex resolution δ
Generate grid $G_\delta \subset \Delta_4$ of candidate strategies
$U^* \leftarrow -\infty$
for each $\boldsymbol{\sigma} \in G_\delta$ **do**
  Solve (4) for $e^*(\boldsymbol{\sigma})$ via bisection
  Compute $U_D(\boldsymbol{\sigma}, e^*(\boldsymbol{\sigma}))$ via (3)
  if $U_D > U^*$ **then**
    $U^* \leftarrow U_D$; $\boldsymbol{\sigma}^* \leftarrow \boldsymbol{\sigma}$
  end if
end for
return $\boldsymbol{\sigma}^*$, $e^*(\boldsymbol{\sigma}^*)$

the attacker's private benefit), weighted by a risk-aversion parameter $\lambda \geq 0$:

$$U_D(\boldsymbol{\sigma}, e) = -\sum_{j=1}^{4} \sigma_j C_j \; - \; \lambda L \sum_{j=1}^{4} \sigma_j \, p_j(e). \tag{3}$$

Equation (3) directly generalizes the three objectives common to cost-driven deployment-selection formulations (minimize deployment cost, maximize attack cost, minimize attack benefit): the first term is the deployment-cost objective, and the second term folds its attack-cost/attack-benefit objectives into a single expected security-loss term whose magnitude the enterprise controls indirectly, via $\boldsymbol{\sigma}$, by shaping the attacker's incentive to invest e.

B. Stackelberg Equilibrium

The attacker's best response solves $\max_{e \geq 0} U_A(\boldsymbol{\sigma}, e)$. Since U_A is concave in e (the benefit term saturates while the cost term $c_1 e^2$ is strictly convex), the best response $e^*(\boldsymbol{\sigma})$ is characterized by the first-order condition

$$B \sum_{j=1}^{4} \sigma_j \theta_j e^{-\theta_j e} - 2c_1 e = 0, \tag{4}$$

which we solve numerically via bisection (the left-hand side is strictly decreasing in e once $e > 0$, guaranteeing a unique root). The enterprise then solves the leader's problem

$$\boldsymbol{\sigma}^* = \arg\max_{\boldsymbol{\sigma} \in \Delta_4} U_D\big(\boldsymbol{\sigma}, e^*(\boldsymbol{\sigma})\big), \tag{5}$$

over the probability simplex Δ_4, which defines the enterprise's Strong Stackelberg Equilibrium (SSE) strategy [3]. This bilevel program has no closed-form solution in general because $e^*(\boldsymbol{\sigma})$ is only implicitly defined by (4); we compute $\boldsymbol{\sigma}^*$ via a discretized simplex search (Algorithm 1), analogous in spirit to the multiple-LP/DOBSS approach used to compute SSEs in classical security games [5], but adapted to our continuous attacker action space.

C. Bayesian Extension: Uncertain Attacker Type

Real enterprises face attackers of heterogeneous sophistication. Following the Bayesian Stackelberg formulation of [5], we extend (2)–(5) to K attacker types $\theta^{(k)} \in \{\text{opportunistic}, \text{advanced}\}$ with prior probabilities q_k and type-specific cost coefficients $c_1^{(k)}$. Each type independently best-responds via (4), and the leader maximizes the prior-weighted expectation $\sum_k q_k \, U_D(\boldsymbol{\sigma}, e^{*(k)}(\boldsymbol{\sigma}))$. This requires no change to Algorithm 1 beyond evaluating the inner loop once per type per candidate $\boldsymbol{\sigma}$, and lets an enterprise reason about, e.g., a mix of opportunistic scanners and resourced/APT-style adversaries without committing to a single worst-case assumption.

TABLE II
SIMULATION PARAMETERS

Architecture j	C_j	θ_j
(i) Physical standalone	100	0.10
(ii) Cloud-native standalone	65	0.30
(iii) Shared RAN + cloud-native	40	0.55
(iv) Dedicated slice	55	0.25

$B = 120$ (attacker benefit scale), $L = 150$ (enterprise loss scale)

V. Evaluation Methodology

Because we have no access to a live non-public 5G testbed, we evaluate SPEAR-5G through a *calibrated numerical simulation* that instantiates Section IV's model with parameters chosen to reflect the qualitative cost and exposure relationships described narratively in [2] (physical standalone deployments are the most expensive but least exposed; shared RAN with shared cloud-native core is the cheapest but most exposed to publicly-reachable infrastructure; dedicated slices recover much of the isolation of standalone deployments at a lower cost). Table II lists the parameters used in all experiments unless otherwise stated. We stress that these are illustrative, simulation-only values, not measurements from a physical deployment; Section VII discusses this limitation further.

We compare three deployment-selection strategies:

- **SPEAR-5G**: the Stackelberg equilibrium $\boldsymbol{\sigma}^*$ from Algorithm 1 ($\delta = 0.05$ simplex resolution).
- **Static baseline**: representative of [2]'s constrained cost-minimization; deterministically selects the single cheapest feasible architecture ($\arg\min_j C_j$), ignoring the attacker's strategic response.
- **Uniform-random baseline**: a naive mixed strategy $\sigma_j = 0.25 \; \forall j$, representing an unoptimized attempt at randomization without game-theoretic reasoning.

For each strategy we compute the attacker's best-response effort via (4) and report the resulting defender total expected cost $\sum_j \sigma_j C_j + \lambda L \sum_j \sigma_j p_j(e^*)$, deployment cost, and expected security loss, sweeping the attacker strength ($1/c_1$), the enterprise's risk-aversion λ, and the degree of strategy randomization (Shannon entropy of $\boldsymbol{\sigma}$, normalized to $[0, 1]$).

VI. Results

Cost vs. attacker strength. Fig. 1 sweeps attacker strength ($1/c_1$, i.e. a cheaper/more capable attacker moving rightward) at fixed $\lambda = 1$. SPEAR-5G's equilibrium strategy achieves the lowest total expected cost across the entire range,

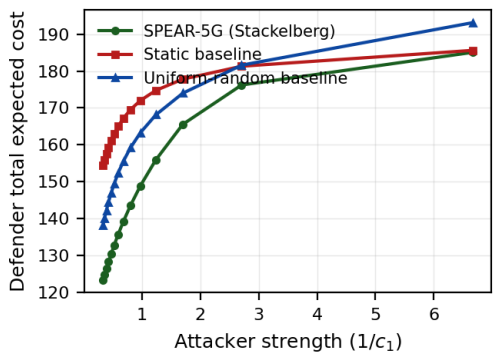


Fig. 1. Defender total expected cost vs. attacker strength ($1/c_1$) for SPEAR-5G, the static baseline, and the uniform-random baseline ($\lambda = 1$).

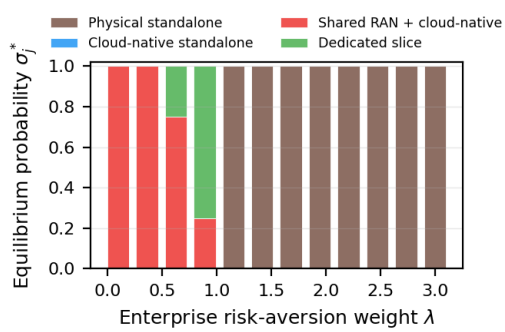


Fig. 2. Equilibrium probability mass on each of the four deployment architectures as a function of the enterprise's risk-aversion λ.

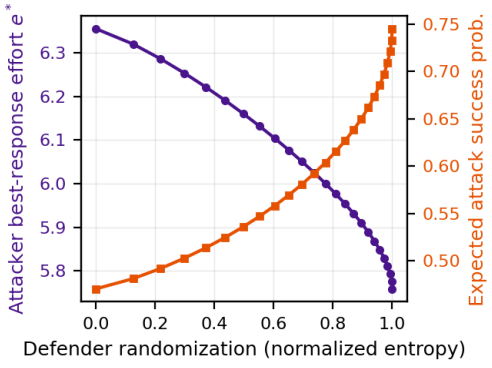


Fig. 3. Attacker best-response effort and expected attack success probability as the defender's strategy is interpolated from the most secure pure architecture toward the uniform distribution.

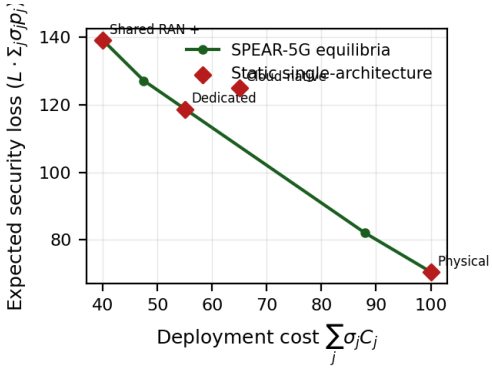


Fig. 4. Pareto frontier of deployment cost vs. expected security loss: SPEAR-5G equilibria (sweeping λ) vs. static single-architecture choices.

with the largest relative gain against the static baseline at moderate attacker strength (up to ≈19% lower total cost), where anticipating the attacker's best response matters most. As attacker strength grows very large, all curves converge, since a sufficiently well-resourced attacker eventually saturates $p_j(e) \to 1$ regardless of architecture.

Equilibrium composition vs. risk-aversion. Fig. 2 shows how the equilibrium mixed strategy shifts as the enterprise's risk-aversion λ increases (fixed attacker strength). For low λ (cost-dominated regime), SPEAR-5G concentrates on the cheap, shared RAN/cloud-native architecture. As λ grows, probability mass shifts first to the dedicated slice (a cost-efficient isolation compromise) and finally to the physical standalone deployment once security dominates the objective – reproducing, as a special case, the qualitative cost-vs-isolation trade-off discussed narratively in [2], but now derived as an equilibrium outcome rather than assumed.

Effect of randomization on the attacker. Fig. 3 interpolates the defender's strategy between the most secure pure architecture and the uniform distribution, and plots the resulting attacker best-response effort e^* and expected attack success probability. Increasing randomization *monotonically reduces* the attacker's concentrated best-response effort e^* (diffusing the attacker's target uncertainty), yet, because *naive* (uniform) randomization spreads probability onto high-exposure architectures, the *expected* success probability can rise even as per-architecture effort falls. This is precisely why the uniform-random baseline underperforms the static baseline in Fig. 1 at low attacker strength: undirected randomization is not automatically beneficial. SPEAR-5G's equilibrium strategy avoids this pitfall by optimizing *where* probability mass is placed (Fig. 2), not merely how much randomization is applied.

Cost–security Pareto frontier. Fig. 4 plots deployment cost against expected security loss, sweeping λ for SPEAR-5G and showing the four static single-architecture choices as discrete points. SPEAR-5G's equilibrium curve lies at or below every static point (coinciding with the static points only at the two extremes, where the optimal mixed strategy degenerates to a pure strategy), i.e. for any deployment cost budget achievable by a static choice, SPEAR-5G achieves equal or strictly lower expected security loss.

VII. Discussion

The results support the paper's central hypothesis: modeling the attacker as a strategic best-responder, rather than a static cost term, changes the optimal deployment decision and improves the achievable cost–security trade-off. Three

implications stand out. First, the gain from SPEAR-5G over the static baseline is not uniform – it is largest in the moderate attacker-strength regime (Fig. 1), suggesting that static, attacker-agnostic cost formulations such as [2] are closest to optimal only at the extremes (a negligible or an overwhelming attacker), and least optimal for the realistic middle ground of moderately resourced adversaries. Second, randomization is not inherently protective (Fig. 3); an enterprise that merely diversifies its deployment mix without game-theoretic reasoning about where exposure is highest can do worse than a deterministic choice, which is an actionable and, to our knowledge, novel observation for the non-public 5G setting. Third, the Bayesian extension (Section IV-C) offers a principled way to hedge against attacker-type uncertainty without resorting to worst-case-only design, echoing established results from the security-games literature [5] transplanted to the 5G deployment-selection problem.

Limitations. Our evaluation is simulation-based with illustrative, hand-calibrated parameters (Table II), not a substitute for measurements on a live testbed such as the one in [2]. The exponential success-probability model (1) and quadratic effort-cost model in (2) are standard security-games modeling choices but are unfit to empirical 5G attack-cost data. The single-shot game also abstracts away attacker learning across repeated interactions, and assumes the attacker observes the defender's mixed strategy, a standard but non-trivial assumption inherited from the classical SSG literature [3].

VIII. Conclusion and Future Work

We introduced **SPEAR-5G**, a Stackelberg security game in which an enterprise commits to a randomized non-public 5G deployment strategy that anticipates a strategically-responding attacker, with a Bayesian extension for attacker-type uncertainty. Simulated evaluations, calibrated to a cost taxonomy consistent with the non-public 5G deployment literature, show SPEAR-5G Pareto-dominates both a static cost-only baseline and a naive uniform-randomization baseline, and surface a non-obvious finding that undirected randomization can be counter-productive. Future work includes: (i) validating the exposure and cost parameters against a live testbed, e.g. a UERANSIM/OpenAirInterface setup; (ii) extending the game to a multi-stage/repeated setting capturing attacker reconnaissance and learning; (iii) incorporating multiple simultaneous attacker types with correlated equilibrium solution concepts; and (iv) coupling SPEAR-5G with reinforcement-learning-based equilibrium computation to scale beyond four architectures to fine-grained per-NF placement decisions.

Disclaimer

This entire manuscript – including the title, framing, literature selection, mathematical formulation, algorithm, simulation code, figures, numerical results, and conclusions – was generated by an AI language model (Claude Sonnet 5, Anthropic) at a user's request, building on the deployment-cost taxonomy and attack model introduced by Tripathi et al. *[2]. The evaluation is based on synthetic, hand-calibrated simulation data chosen to be qualitatively plausible, not on measurements from a real non-public 5G deployment or testbed. The reference list was checked against web search results for bibliographic accuracy at generation time, but neither the technical contribution nor the results have been peer-reviewed or validated by a human domain expert, and the paper should not be treated as a genuine scientific contribution, cited as prior work, or used as a basis for real deployment decisions without independent verification. No human author, affiliation, or institution is associated with this document.*

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